Week #11 : Membrane Transport — GRASPS Performance Task and Practice Questions

Subject/الموضوع: AP Biology Unit Title: Cells / Membrane Transport Grade12: الصف

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can predict how water potential and solution concentration influence osmosis. I can conduct and record two virtual osmosis trials using original evidence.	1.1 Flexible display of information (size, color, font, layout).	Grade 12 Membrane Transport GRASPS task, pp. 1–2; ENS virtual lab (linked in task; requires accompanying HTML file); individual investigation sheet, device, screenshots, notebook.
Lesson 2 الحصة الثانية	I can compare passive diffusion with ATP-dependent active transport using virtual trials. I can organize results, graph osmosis data and recommend a suitable solution using evidence.	5.3 Build fluency with graduated scaffolds and models.	GRASPS task, pp. 1–2; virtual lab; original trial tables at 0, 2, 4 and 6 minutes; graph paper/spreadsheet; evidence briefing slides.
Lesson 3 الحصة الثالثة	I can communicate an evidence-based recommendation about membrane transport. I can justify my conclusions and evaluate two limitations of the simulation during an individual scientific defense.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	GRASPS task and 20-mark regular Biology rubric, pp. 3–4; 6–8-slide briefing; investigation record, graphs, screenshots and individual presentation.
Lesson 4 الحصة الرابعة	I can apply membrane structure and transport knowledge to multiple-choice practice questions. I can explain and correct my answers using scientific evidence and appropriate terminology.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Uploaded Membrane Transport Practice Questions PowerPoint: 25 self-assessment questions and answer key; notebook; individual corrections and peer discussion.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	• Lessons 1–3: GRASPS project is individual; retain original virtual-lab readings, screenshots, graph, and short oral defense. • The project requires the accompanying HTML lab file to be accessible alongside the Word task. • Lesson 4: use the uploaded 25-question PowerPoint for practice and corrections. • Project rubric: four categories, 5 marks each (20 marks total).		

الأسبوع# Week #11

Technology Integration دمج التكنولوجيا	• Membrane transport virtual lab for osmosis, diffusion and ATP trials (Lessons 1–2). • PowerPoint for 6–8-slide evidence briefing (Lesson 3). • Uploaded practice-question PowerPoint and answer key (Lesson 4); Microsoft Teams for sharing resources.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: plotting cell-water readings over time, comparing numerical changes, interpreting graphs. • Chemistry: solute concentration, gradients and water potential. • Health sciences: solution selection for a simplified model of cell storage (not clinical testing).
External Assessment Integration دمج التقييمات الخارجية	• AP Biology Unit 2: Cell Structure and Function — membrane permeability, passive transport, osmosis, water potential, and active transport. • Lesson 4: 25 membrane structure/transport self-assessment questions with justification and error correction. • GRASPS performance task uses the uploaded 20-mark regular Biology rubric.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: predict, analyze quantitative trends, evaluate model limitations and defend a recommendation. • Research: collect original virtual-lab evidence and organize tables/graphs. • Communication: concise briefing and individual scientific question. • Self-management: complete three staged project deadlines and reflect on corrections.
Standards المعايير	• Lesson 1: AP Biology Unit 2 — osmosis, water potential, and concentration differences across membranes. • Lesson 2: AP Biology Unit 2 — simple/facilitated diffusion, active transport and ATP use; data analysis. • Lesson 3: AP science practices — evidence-based explanation, interpretation and communication. • Lesson 4: AP Biology Unit 2 — membrane structure and transport review.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Lesson 1: How does changing the surrounding solution affect the direction of water movement? Lesson 2: Why does ATP availability change the outcome of transport against a gradient? Lesson 3: What evidence is sufficient to recommend conditions for a model cell, and what are the limitations? Lesson 4: How can we distinguish between transport mechanisms in unfamiliar situations?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — GRASPS: Predict & Osmosis MUST (DOK 1): Identify direction of osmosis and record two trials. SHOULD (DOK 2): Compare cell-water changes at 0, 2, 4 and 6 minutes. COULD (DOK 3): Explain both patterns using water potential. Lesson 2 — GRASPS: Diffusion & ATP MUST (DOK 1): Complete one diffusion and two ATP trials. SHOULD (DOK 2–3): Graph and interpret osmosis results; compare ATP on/off. COULD (DOK 4): Justify a solution recommendation using data and model limitations. Lesson 3 — GRASPS: Present & Defend MUST (DOK 1): Present original results and a clear recommendation. SHOULD (DOK 2–3): Defend claims with readings and explain two limitations. COULD (DOK 4): Evaluate trade-offs and respond independently to a scientific question.

	Lesson 4 — Practice Questions MUST (DOK 1): Identify membrane structures and transport methods. SHOULD (DOK 2–3): Justify answers and correct misconceptions. COULD (DOK 4): Evaluate unfamiliar membrane-transport scenarios.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Innovation الابتكار
	المجال Area: Culture / Innovation / Sustainability
	• Link virtual modeling to responsible UAE innovation in laboratory and biotechnology education. • Discuss the value and limitations of using models before making decisions relevant to healthcare and cell preservation. • Connect evidence-based problem solving with the UAE's emphasis on scientific innovation.